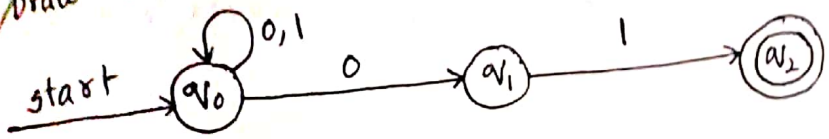
Draw transition table

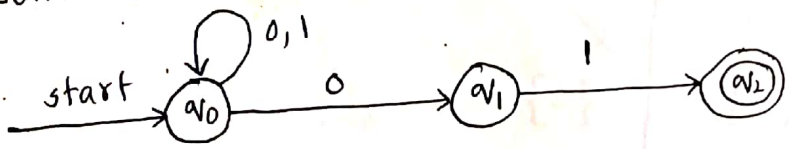


Ans:

	0	1
→ q_0	$\{q_0, q_1\}$	$\{q_0\}$
q_1	$\emptyset$	$\{q_2\}$
* q_2	$\emptyset$	$\emptyset$

Convertion of NFA to DFA

Convert the following NFA to DFA



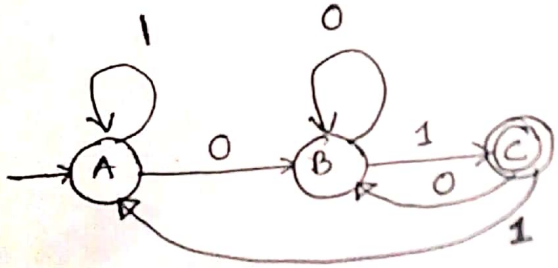
Ans:

	0	1
→ $\{q_0\}$	$\{q_0, q_1\}$	$\{q_0\}$
$\{q_0, q_1\}$	$\{q_0, q_1\}$	$\{q_0, q_2\}$
* $\{q_0, q_2\}$	$\{q_0, q_1\}$	$\{q_0\}$

⇓

	0	1
→ A	B	A
B	B	C
* C	B	A

⇒



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Exercise:

Give DFA's accepting the following languages over the alphabet $\{0, 1\}$

a) The set of all strings ending in 00.

b) The set of all strings with three consecutive 0's (not necessarily at the end).

c) The set of strings with 011 as a substring

Exercise

Convert the following NFA into DFA

a)

	0	1
→ p	{p, q}	{p}
q	{r}	{r}
r	{s}	∅
*s	{s}	{s}

b)

	0	1
→ p	{q, s}	{q}
*q	{r}	{q, r}
r	{s}	{p}
*s	∅	{p}

Application of Finite automation:

- Web search
- Information retrieval from text.

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